ASSIGNMENT 9

1.
Create an index that will enable a user to pull orders for ‘1990-10-03’ out of the Orders table quickly.

```
KD3-NehaDeshmukh-96628>CREATE INDEX idx_orders_odate ON orders (odate)
Query OK, 0 rows affected (0.12 sec)
Records: 0  Duplicates: 0  Warnings: 0

KD3-NehaDeshmukh-96628>SELECT * FROM orders;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
10 rows in set (0.00 sec)
```

2.
If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
KD3-NehaDeshmukh-96628>alter table  orders add constraint uniq_onum UNIQUE(onum)
;
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
KD3-NehaDeshmukh-96628>SHOW INDEX FROM orders;
+--------+------------+-----------------+--------------+-------------+---------
--+-------------+----------+--------+------+------------+---------+-------------
--+---------+------------+
| Table  | Non_unique | Key_name        | Seq_in_index | Column_name | Collatio
n | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_commen
t | Visible | Expression |
+--------+------------+-----------------+--------------+-------------+---------
--+-------------+----------+--------+------+------------+---------+-------------
--+---------+------------+
| orders |          0 | uniq_onum       |            1 | onum        | A
  |          10 |     NULL |   NULL | YES  | BTREE      |         |
  | YES     | NULL       |
| orders |          1 | idx_orders_odate |           1 | odate       | A
  |           2 |     NULL |   NULL | YES  | BTREE      |         |
  | YES     | NULL       |
+--------+------------+-----------------+--------------+-------------+---------
--+-------------+----------+--------+------+------------+---------+-------------
--+---------+------------+
2 rows in set (0.03 sec)
```

3.Create an index that would permit salesperson to retrieve his orders.

```
KD3-NehaDeshmukh-96628>CREATE index indx_order_snum on orders (snum);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

KD3-NehaDeshmukh-96628>SHOW CREATE TABLE orders;
```

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

```
KD3-Prerana-96192>CREATE INDEX i_c on customers(snum,rating);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

5.Create an index to speed up searching order on a given date by given customer.

```
KD3-NehaDeshmukh-96628>create index idx_order_odate__cnum on orders (odate, cnum
);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```